Pulling the Right Lever: School Attainment, Open Data Analytics and Local Education Policy in England

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Abstract

The UK government's 2026 White Paper targets halving the disadvantage attainment gap. Achieving this requires local education authorities to understand the drivers of attainment in their areas, but challenges in evidence synthesis and locally relevant data analytics mean the risk of pulling the wrong policy lever is high. Using linear mixed-effects models fitted to 4-years of DfE data, we show school-level Attainment 8 is highly predictable. Focusing on Brighton and Hove, where 2024 secondary admissions changes were premised on the idea that disadvantaged student attainment could be improved through even more social mixing in the school system, our analysis suggests that through omitted variables, non-linearities and counter-intuitive relationships, the policy is unlikely to deliver the targeted gains, while the wider literature on long school commutes points to social and educational costs not adequately considered. The city has the second-worst absence rate in England in 2024-25 (and among the worst throughout the period), and our modelling indicates improving attendance would be the most impactful single policy lever, particularly for disadvantaged pupils. We also introduce a school-level value-added view that lets parents and policy makers assess how a school performs above its inherited intake characteristics, and a policy simulator tool to bridge the gap between open data and accessible, contextualised intelligence.

1 Introduction

Pupil outcomes, including attainment, are a recurrent policy theme given the established links to individual life outcomes (Farquharson et al. 2024) and our collective national success through measures such as economic productivity (Grant 2017). The UK Government's 2026 schools white paper (DfE 2026) sets ambitious targets including halving the disadvantage attainment gap, improving attendance, reforming admissions and tackling place-based disadvantage. These ambitions will require local education authorities (LEAs) to understand the specific drivers in their areas and choose interventions accordingly.

Focusing on the secondary school attainment, this paper argues that the Department for Education's (DfE) open data provides most of the raw materials needed for such nuanced localised understanding, but a crucial gap exists between data availability and accessible, contextualised intelligence for decision makers. We demonstrate this through an analysis of school-level attainment across England, combined with a case study of Brighton and Hove where a policy direction would be predicted by our model, to potentially do more harm than good.

Brighton and Hove provides a particularly instructive case. The city already has a distinctive admissions landscape featuring catchment areas with a lottery tie-break for oversubscribed schools, introduced following a secondary school closure in 2005 (Allen et al. 2013), and an admissions priority for Free-School-Meal (FSM)-eligible pupils across its community schools. In 2024, the council launched proposals to reduce Published Admission Numbers at popular central-catchment schools, reserve places for out-of-catchment children in those catchments ahead of in-catchment children, and redraw catchment boundaries. The proposals were promoted on the premise that attainment in the city was 'driven by economic advantage' and that redistributing pupils would narrow the attainment gap. The public response was overwhelmingly negative, yet parents and schools lacked the tools to challenge the proposals effectively and the core policy direction persisted, albeit slightly watered down.

Raw data on the attainment gap, without context, was presented to the city alongside statements correlating concentrations of disadvantage with lower attainment. The gap was framed as a unique failure requiring urgent action; the second-order impacts of the proposed interventions - longer school journeys, displacement from local communities, effects on well-being - were not scrutinised in depth. Our modelling addresses a critical blind spot in local decision-making: without the ability to account for structural factors like absence and prior attainment, councillors and citizens were unaware that, *after controlling for absence in particular*, Brighton and Hove was already among the highest-performing LEAs nationally for taking students beyond those factors - ranking 15th of 151 for disadvantaged pupils and 4th for non-disadvantaged pupils. As we show later, that conditional ranking is itself a measure of how much the city's severe absence problem is costing its disadvantaged pupils: set absence aside and the schools look exceptional; leave it in the ledger and the city is ordinary. These insights come from the analysis of open data presented in this paper and carried out after the consultation had closed. However, with more timely analysis and with this knowledge, the approach to improving outcomes for disadvantaged pupils in the city could have been more effective and different questions asked: "what can we learn from what the city is doing right?" and "would asking pupils to travel out of their communities to schools jeopardise this success?" Some rapidly-produced analysis was put into the public domain by some of the authors of this paper[1], but as a response to, not a contribution to, policy formation.

The discussions that led to this paper took place over an 18-month period spanning the 2024 and 2025 admissions consultations and involved a diverse group of parents, academics, policy practitioners and school governors. We do not claim our model is optimal or novel - the variables and effects are very similar to those reported by Macleod et al. (2015) over a decade ago. Our intention is to show that with freely available open data and a small set of well-chosen variables, it is possible to build a model explaining close to 80% of the variation in school-level Attainment 8, and then to use it to answer locally targeted questions such as: 'is changing the social mix in schools likely to improve disadvantaged pupil attainment?', 'what is the most important factor affecting attainment in a particular LEA?', and 'which schools are doing better or worse than expected?'. We report effects in GCSE-point units rather than standard deviations so that they have more direct meaning for policy makers.

Before examining the Brighton and Hove proposals specifically later in the paper, we wish to be explicit about the scope of our argument. Our models measure attainment outcomes only; they cannot speak to the broader social, civic and developmental value of diverse school communities, which we do not dispute. We are also aware that our findings regarding concentrations of disadvantaged students could be read selectively to justify resistance to integration efforts in general — that is not our argument. Our claim is specific: in this city, at this time, with this evidence base, the proposed mixing intervention was not the most impactful lever available for disadvantaged student attainment, and carried costs the consultation did not adequately weigh.

The paper proceeds as follows. We briefly review the literature on secondary school attainment drivers, then outline our modelling approach using DfE open data. We present results showing which factors matter most and how they interact, before applying these findings to Brighton and Hove. We conclude with implications for local and national education policy.

2 Literature review: factors affecting pupil attainment

2.1 Socio-economic background and disadvantage

The Education Policy Institute's annual reports document that by the time pupils sit their GCSEs (the General Certificate of Secondary Education - the exam most students in England sit at age 16), disadvantaged pupils are on average around 18 months of learning behind their more affluent peers, a gap that narrowed modestly after 2011 but partially reversed following the COVID-19 pandemic (Tuckett et al. 2023). It is this 'attainment-gap' which has remained stubbornly persistent for many years and that the recent government white paper has resolved to try and tackle. The size of the gap and the reasons behind it are complex and vary from place to place, but its existence has spawned large volumes of literature delving into disadvantage and attainment (for wider reviews and evidence syntheses, see e.g. Stopforth

[1] https://osf.io/x82j3/overview (see 'about.html' in 'supplementary.zip')

and Gayle (2025); Macleod et al. (2015); Tuckett et al. (2023)).

Free School Meal (FSM) eligibility serves as the standard proxy for disadvantage, though its limitations are well established. For example, Stopforth and Gayle (2025) demonstrate that finer-grained measures of parental social class reveal persistent attainment gaps that FSM-based analyses tend to understate, with the thresholds for FSM eligibility low enough that many students who might otherwise be classed as disadvantaged, are not included in these statistics. Despite this, because eligibility for Free School Meals is regularly recorded at pupil and school levels, it is frequently used as a proxy.

Inequalities in GCSE outcomes are substantially mediated by prior attainment at National Curriculum Key Stage 2 (KS2 - the latter stages of primary school from ages 7 to 11[2]), reflecting accumulated inequalities from before secondary school (Stopforth and Gayle 2025). Gorard and Siddiqui (2019) show that the duration of disadvantage is a stronger predictor than disadvantage at any single point, and propose that pupils do worse in schools with higher concentrations of disadvantage, estimating improvements of 0.05–0.15 standard deviations (approximately one GCSE grade in one subject) from greater social mixing. They argue this can be achieved at 'almost no cost' - a conjecture that overlooks real-world consequences for communities, particularly where local social geographies mean de-concentration requires children travelling long distances.

2.2 Attendance

School attendance is among the strongest predictors of attainment and this comes through strongly in the literature. DfE analysis of 2018/19 data (DfE 2022) documents a stark gradient: among persistently absent pupils, only 35.6% achieved a standard pass in English and Maths, compared with 83.7% among those with no recorded absences. It is also cited as one of the main factors in pushing down attainment for disadvantaged pupils by Macleod et al. (2015). More recent analysis finds that pupils with near-perfect attendance were 1.9 times as likely to achieve a grade 5 pass compared with otherwise similar pupils attending 90–95% of the time (DfE 2025a). Dräger et al. (2024), using the 1970 British Cohort Study, find that absences at age 10 are associated with lower educational attainment in adulthood. The Social Mobility Commission (Riordan et al. 2021) conclude that the correlation between attendance and progress is 'most likely to be causal'.

Post-pandemic persistent absence has risen sharply, from around 10.5% pre-pandemic to approximately 21.9% in secondary schools by 2023/24, with rates for FSM-eligible pupils around 32% (DfE 2025b). This compounds the challenge of disentangling school-level from pupil-level effects.

How far attendance is amenable to *school-level* action is an important question. Some of the variation reflects school practice - pastoral systems, attendance follow-up and ethos - but a substantial share is driven by circumstances beyond the school gate: family poverty, physical and mental ill-health, special educational needs, caring responsibilities, and the broader adversities that depress attainment through multiple channels (Houtepen et al. 2020; Rakesh

[2] Details of English National Curriculum Key Stages https://www.gov.uk/national-curriculum

et al. 2025). The steep, largely exogenous rise in absence since the pandemic underlines how much of the recent problem sits outside any individual school's control. This distinction - between the intake-driven component of absence and the smaller portion a school can itself move - is central to how absence should be interpreted as a policy lever, and we make it explicit in our modelling in section 5.5. It also implies that authority-wide improvement is unlikely to come from school-led practice-sharing alone, and more probably requires strategic, council-led, cross-departmental action spanning health, children's services and family support alongside schools.

2.3 Other factors

Ethnic inequalities are persistent (Gillborn et al. 2017), though Gorard et al. (2025) show these are substantially reduced when controlling for pupil-level characteristics. The under-performance of disadvantaged White British pupils, particularly boys, has attracted policy attention, with causes contested between place-based deprivation and cultural factors (Strand 2021; Commons Education Committee 2021).

School-level selectivity inflates raw performance metrics. Anders et al. (2024) find that the private school premium effectively disappears once socio-economic background is controlled for. Burgess et al. (2018) show grammar schools reinforce socio-economic stratification without overall attainment benefits.

Workforce factors including effective leadership (Zuccollo et al. 2023), teaching quality (Burgess et al. 2023), and teacher retention (Gibbons et al. 2018; Menzies 2023) contribute to variation in attainment, with effects felt most acutely by disadvantaged students (Allen et al. 2018). The 'London Effect' (the apparent boost to attainment for disadvantaged pupils who live in the Capital) (Ross et al. 2020) reflects a localised concentration of agency factors - parental expectations, homework hours, lower unauthorised absence - rather than a purely geographic phenomenon.

The factors affecting attainment are multifaceted and interrelated, and many drivers are well-established. With so many concurrently relevant predictors, however, it is difficult to judge their relative importance in any specific context. What is required is an analysis combining as many drivers as possible, so that their relative influences, interactions, and mediation pathways can be assessed concurrently - which is where we now turn.

3 Data and methods

3.1 Data

The DfE publishes extensive open data on schools via its statistical services. We assembled a panel dataset covering the four post-COVID academic years (2021–22 to 2024–25), linking school performance, absence, pupil population, workforce and Ofsted data. Data pre-processing details are available in the supplementary material.[3] The full panel comprises 13,419 school-

[3] Supplementary material: https://osf.io/x82j3/overview (see 'index.html' in 'supplementary.zip') - with links through to the source code on Github.

year observations from 3,523 academies and maintained schools across 152 LEAs. Independent schools, colleges and 'special schools' (those catering more for pupils with special educational needs and disabilities) are excluded. After the exclusion of cases with missing predictor values or imputation of those values required for the log-linear specification (principally missing workforce or Ofsted data in 2024-25, together with KS2 prior attainment, which is not published for that year because the relevant cohort's primary assessments were cancelled during the pandemic - see section 3.2[4]), the analysis sample used in the models is approximately 12,200 observations. All data are school-level aggregates; interpretations relate to this level of aggregation.

3.2 Model specification

We fit multilevel linear mixed effects models (LMEs) with schools nested within LEAs within regions (Snijders 2012). The outcome is log-transformed Attainment 8 (for all, disadvantaged and non-disadvantaged pupils), the measure of interest cited in the DfE (2026) white paper. We have selected Attainment 8 over Progress 8 as we want to assess the influence and interaction variables like prior attainment on other predictor variables, not just as a partial outcome. Because the outcome is on the log scale, coefficients for log-transformed predictors represent elasticities. We should be clear that the log transformation is *not* applied uniformly to every predictor: only the four skewed, strictly-positive count-and-rate variables - the disadvantage, absence and EAL percentages and teacher sickness days - enter in logged form, chosen on the basis of visual fit and the interpretability of elasticities. The remaining predictors, including mean KS2 prior attainment, the segregation index and the workforce proportions, enter linearly. This variable-by-variable choice, together with supporting diagnostics (component-plus-residual plots and a comparison of logged, linear and spline treatments), is documented in the supplementary material.

The model specification is:

$$\log(\text{ATT8}_{ij}) = \beta_0 + \sum_{k=1}^{9} \beta_k \, x_{kij} + u_{\text{year}} + u_{\text{Ofsted}} + u_{\text{region}} + u_{\text{LA}|\text{region}} + \varepsilon_{ij}$$

where i indexes schools, j indexes year-observations, the β_k are fixed-effect coefficients, and the u terms are random-effect intercepts for year, Ofsted rating, region, and LEA nested within region.

Fixed-effect predictors comprise: log(% KS4 pupils who are disadvantaged - % FSM), log(% overall absence - % Absence), log(% pupils with English as an additional language - % EAL), mean KS2 prior attainment (the cohort's average KS2 scaled score, centred at the national standard of 100), selective admissions (dummy), Gorard Segregation Index (LA-level), teacher retention rate, leadership pay proportion, and log(teacher sickness days).

The selective-admissions variable is the DfE's own performance-tables admissions classification, and within our state-funded sample it distinguishes three groups: wholly *selective* state

[4] missing data and imputation: https://osf.io/x82j3/overview (see 'data_overview.html#missing-data' in 'supplementary.zip')

grammar schools that admit by an entrance test (166 schools, 4.8% of the sample); non-selective schools located in otherwise selective areas (234 schools, the reference category - we note for those not familiar with the English system that following reforms some decades ago, only some LEAs in England still have selective grammar schools); and all other non-selective schools (the remaining ~88%). The estimate we report is therefore the effect of *full* academic selection, benchmarked against non-selective schools whose intakes are shaped by the same local grammar systems. It does not capture partial selection, banding, aptitude-based selection or faith-based oversubscription criteria, all of which fall within the 'other non-selective' category and are consequently unmeasured here; faith schools in particular are not separately identified in the main model.

There are two points relating to the prior-attainment variable that need highlighting. Firstly, we control for prior attainment using the cohort's *mean KS2 scaled score*, and for the final year of our panel no observed prior-attainment data exists at all. Because the KS2 tests and assessments of 2019-20 and 2020-21 were cancelled during the pandemic, the cohorts reaching the end of KS4 in 2024-25 and 2025-26 have no KS2 results of any kind; the DfE accordingly publishes neither prior-attainment measures nor Progress 8 for those years.[5] For 2024-25 we therefore carry forward each school's most recently observed value (or, where that is unavailable, its mean across observed years), and the affected rows are flagged in the panel (see Supplementary material[6] for methods and diagnostic plots). Secondly, because we lack pupil-level data, prior attainment is corrected only at the school level; this is a coarser adjustment than a pupil-and-school-level correction would provide, and therefore we can't rule out some confounding of the intake coefficients where a common cause exists. We return to this limitation later in the paper.

An earlier version of this analysis controlled for prior attainment using only the *percentage of pupils with low KS2 prior attainment*. In this iteration we look at the mean KS2 score. This implications of this change are discussed below.

We fit separate models for overall Attainment 8, disadvantaged-pupil Attainment 8, and non-disadvantaged-pupil Attainment 8, as both panel models (all four years) and individual year models. Models were estimated using the `lme4` package in R with Restricted Maximum Likelihood estimation.

Variable selection was informed by the literature review and by the policy narrative in Brighton and Hove, where segregation featured prominently in the council justification. Full details of the variables and the full analysis panel can be found in the supplementary material [7].

[5] Department for Education, *Key stage 4 performance*: "It is not possible to calculate Progress 8 for academic years 2024/25 and 2025/26. This is because there is no KS2 prior attainment data available to use to calculate Progress 8 when the relevant cohorts reach the end of KS4 as primary tests and assessments were cancelled in academic years 2019/20 and 2020/21 due to COVID-19 disruption." https://explore-education-statistics.service.gov.uk/find-statistics/key-stage-4-performance

[6] See data overview for details https://osf.io/x82j3/overview (see 'data_overview.html' in 'supplementary.zip')

[7] See data overview for details https://osf.io/x82j3/overview (see 'data_overview.html' in 'supplementary.zip')

4 Results

4.1 Model fit

The models explain a substantial proportion of variance in school-level Attainment 8. In order that these fits are more intuitive for those used to interpreting OLS regression models, we use the method described by Nakagawa and Schielzeth (2013) and Nakagawa et al. (2017) to calculate indicative R^2 values proxying the proportion of Attainment 8 variance explained by each model. For all pupils, the conditional R^2 (fixed plus random effects) is 0.78, with marginal R^2 (fixed effects alone) of 0.65. For disadvantaged pupils the conditional R^2 is 0.73; for non-disadvantaged pupils, 0.8. School-level attainment is remarkably predictable from these structural factors, which makes them very useful for exploring the effectiveness of local policy levers.

4.2 Stepwise decomposition

[Table 1: Stepwise progression to the full multilevel model for overall Attainment 8. — about here]

The stepwise progression in Table 1 reveals the mediating structure among the key predictors. In Model 1, we start with the % KS4 pupils who are disadvantaged (%FSM for short) as the only predictor - mainly as this was the variable that has dominated the policy discussion in Brighton and Hove in recent years, leading to the 2023 FSM priority policy and the 2024 mixing policy. the FSM coefficient is –0.201 and statistically significant, on its own accounting for almost 40% of the variation in Attainment 8 across schools in England. However, adding absence in Model 2 halves it to –0.122. Absence shares explanatory power attributed to deprivation, reducing the coefficient of %FSM, and adds further explanation, together they now account for over 60% of the variation in Attainment 8 scores. Adding prior attainment in Model 3 reduces both FSM and absence coefficients further. By Model 3, with just three variables, close to 70% of attainment variation is explained.

Ranking these three predictors by importance needs more care than it might appear, and it is worth being explicit about why. The t-values in Model 3 (% absence -60.5, mean KS2 58.2, % FSM -21.0) describe how precisely each coefficient is estimated once the other two are held constant — that is, each variable's *unique* contribution. But these three variables are strongly correlated with one another (% FSM with mean KS2 at -0.69, % absence with mean KS2 at -0.59), so a great deal of the variance they explain is shared between them rather than belonging to any one. A variable that shares heavily with the others will have a modest t-value even where it accounts for a large share of attainment on its own, and t-values therefore cannot settle which factor matters most.

Two measures that can apportion the shared variance point the same way as each other. Taken on its own, mean KS2 explains 59% of the variation in Attainment 8, absence 50% and % FSM 39%. Apportioning the shared variance more formally, by averaging each variable's contribution across every possible order in which the three could enter the model (the Shapley or LMG decomposition (Grömping 2007)), gives mean KS2 42% of the explained variance, absence 36% and % FSM 23%.

Prior attainment, then, is the largest single source of variation in school-level attainment, with absence close behind — consistent with the standardised coefficients we report in section 4.3. The point that matters for policy survives every one of these measures: **concentrations of disadvantage come last on all three**, whether judged by unique contribution, by explanatory power alone, or by apportioned share. We return in section 4.3 to what follows from this, and in particular to the distinction between a factor that is large and a factor that can be acted upon.

Before we get to Model 4, at this point, it is worth noting that in this iteration of the analysis we switch from low KS2 prior attainment to mean KS2 score - the difference materially affects two of our estimates. The low-attainment share measures only the bottom of a school's intake distribution: it records how many pupils arrived already behind, but says nothing about how many arrived ahead. Two schools with identical proportions of low prior attainers can have very different intakes across the rest of the attainment spectrum. A model controlling only for the low-attainment share cannot distinguish them, and so systematically under-adjusts for schools whose advantage lies at the *top* of the distribution. Mean KS2 score, contrastingly, reflects the whole distribution - both tails - and produces a better model fit (a reduction in AIC of over 500 for the all-pupil model); once it is included, the low-attainment share adds nothing further (t = -0.45).

This matters for the interpretation of the analysis. The coefficient on concentrations of disadvantage falls by roughly a *quarter* for all pupils, because the low-attainment share had been leaving part of the intake difference between high- and low-FSM schools unmeasured, and the FSM variable had been absorbing it. Properly controlled, disadvantage concentration is an even *weaker* predictor of attainment than our earlier specification implied. For disadvantaged pupils specifically, the positive coefficient on disadvantage concentration becomes both larger and far more clearly distinguishable from zero (t rising from 2.1 to 5.6) - disadvantaged pupils in high disadvantage schools are even more positively affected by this than our earlier specification led us to believe. The absence coefficient, by contrast, barely moves - under 3% in either pupil group - so the paper's central finding is unaffected by the choice. The practical lesson is that in modelling KS4 outcomes it is not sufficient to ask how many pupils arrived behind; how many arrived ahead matters just as much, and controlling just for the low prior attainment tale has the effect of quietly over-crediting schools for intake advantages they did not create.

We should add a note of caution relating to interpretations following this improvement. Prior attainment at Key Stage 2 is not independent of disadvantage: a large chunk of prior attainment actually reflects the inequalities a child has already experienced before secondary school ever begins (Stopforth and Gayle 2025; Gorard and Siddiqui 2019). A variable that captures the whole intake distribution therefore absorbs correspondingly more of the poverty signal than one that captures just the low attainment tail - which is precisely why the disadvantage coefficient attenuates when we make the switch from low to mean KS2 attainment. The coefficient that survives this control should not, then, be read as an estimate of how much poverty matters to attainment. It is a *within-intake* comparison: what remains once schools receiving similar cohorts are set alongside one another. The total contribution of disadvantage to attainment is substantially larger than that residual coefficient implies, and nothing in our analysis should be taken to suggest otherwise. What our models do identify is the marginal association with the *concentration* of disadvantage at school level, holding intake constant - a much narrower quantity, and the one the 2023 (FSM priority) and 2024 (increased mixing) Brighton and Hove policy propositions actually turned on.

Prior attainment carries an additional structural meaning due to Ofqual's 'comparable outcomes' approach, which calibrates national GCSE grade boundaries against the incoming cohort's KS2 profile. Because this calibration is a national, year-to-year adjustment, any mechanical scaling is absorbed by our year random effect. The fixed effect for mean KS2 therefore safely isolates the genuine within-year, between-school association between inherited intake and attainment.

Returning to the stepwise decomposition, Model 4 adds additional fixed effects including EAL proportions, selective admissions, workforce variables and the Gorard Segregation Index. Model 5 adds random effects for year, Ofsted rating, region and LEA nested within region, lifting the R^2 from 0.65 (marginal) to 0.78 (conditional), indicating that 13% of attainment variance is tied to these grouping factors - capturing unmeasured factors such as school culture, leadership quality, local authority support services, and community characteristics. The marginal R^2 captures the model's total explained variance, which includes the substantial variance shared between correlated structural predictors due to shared underlying causes (e.g. poverty). As Table 1 demonstrates, when new variables enter the model, this shared variance is absorbed into the overall predictive power of the model rather than being credited to any single predictor. Consequently, the individual coefficients shrink, leaving them to represent strictly unique, partial effects to compare.

Ofsted rating has only four ordered categories, fewer than would ordinarily be modelled as a random effect. We include it as a grouping factor for partial pooling and consistency with the other contextual groupings, but we have verified that treating it instead as a categorical fixed effect leaves every other coefficient unchanged to three or four significant figures and does not move Brighton and Hove's rank (supplementary material[8]). The fixed-effect form is also informative in its own right: relative to an 'Outstanding' school, and holding intake constant, a 'Good' rating is associated with roughly 1.9 fewer Attainment 8 points, 'Requires Improvement' with 4.0 fewer, and 'Inadequate' with 4.5 fewer - a scale that flattens markedly at the bottom, so that the gap between the two lowest ratings is small. We treat Ofsted as a contextual control rather than a causal factor throughout, since inspection judgements are formed with sight of a school's results and are therefore partly an expression of attainment rather than an independent driver of it; consistent with this, omitting Ofsted entirely *increases* the estimated absence effect by 12–15%, so its inclusion makes our headline absence figure, if anything, conservative.

The Gorard Segregation Index is not statistically significant in Models 4 or 5. Once a school's own FSM levels, attendance, prior attainment and geographic location are accounted for, the distribution of disadvantaged students across schools within an LEA adds no predictive power. Any harm that segregation causes is fully accounted for by the variables already in

[8]See supplementary material for robustness checks: https://osf.io/x82j3/overview (see 'model_experiments.html' in 'supplementary.zip')

the model. We include it because segregation arguments formed much of the justification for the Brighton and Hove admissions proposals.

A crucial feature of these models is that the log-transformed variables encode non-linear relationships. The same percentage-point change produces different effects depending on where a school starts. For concentrations of disadvantage, the relationship between FSM and attainment flattens above about 20% FSM - further increases in disadvantage do not correlate with noticeable further declines. Most of the strong effect appears below 15%, where further reductions are correlated with much steeper attainment increases. This non-linearity has direct policy consequences: individual LEAs need to consider carefully where each school sits on the distribution to appreciate what changes might realistically achieve. We make the practical magnitude concrete in section 5.3 (Figure 4), which translates the coefficients into the expected Attainment 8 gain from a one-percentage-point change at each point along the absence and disadvantage distributions.

4.3 Relative variable importance

[Figure 1: Standardised coefficients showing relative variable importance across pupil groups. Bars show the change in log(ATT8) associated with a one-SD shift in each predictor. — about here]

Figure 1 compares the standardised coefficients[9] across all three pupil groups (all pupils, disadvantaged pupils and non-disadvantaged pupils - full model outputs available in the supplementary materials[10]). Two predictors stand well clear of the rest, and which of them leads depends on the pupil group. Mean KS2 prior attainment has the largest standardised coefficient for all pupils and for non-disadvantaged pupils; for disadvantaged pupils, absence is larger. Prior attainment enters with a *positive* sign - cohorts arriving from primary school with higher average KS2 scores go on to higher Attainment 8 - and its prominence is unsurprising, since it is the closest thing in the model to a direct measure of the intake a school receives.

The distinction that matters for policy is that these two variables are not equally available to act on. Prior attainment is inherited: for a cohort already in secondary school it is fixed, and no admissions or improvement decision taken by a local authority can alter it. It belongs in the model as a control for intake, not as a lever. Absence, by contrast, is at least partly open to intervention, and it is the largest such factor for every pupil group. It is also markedly more important for disadvantaged pupils; for this group, the standardised coefficient for absence is roughly a third larger than the next most impactful predictor, whereas for non-disadvantaged pupils, prior attainment is the dominant factor.

Figure 1 makes the relative importance of the predictors explicit, and the ordering is the main empirical take-home message of the paper. Putting prior attainment to one side - as it

[9] Standardised coefficients are computed as $\beta_k^* = \hat{\beta}_k \times \mathrm{SD}(x_k)/\mathrm{SD}(\log \mathrm{ATT8})$, where $\mathrm{SD}(x_k)$ is the sample standard deviation of the predictor (log-transformed where applicable). This rescales each coefficient to represent the change in outcome, in standard-deviation units, associated with a one-standard-deviation change in the predictor.

[10] Stepwise Models for all pupil groups https://osf.io/x82j3/overview (see ‘model_results.html#stepwise-model-building‘ in ‘supplementary.zip‘)

describes the intake rather than anything a school or authority can change - school absence is the dominant lever by a clear margin; and the concentration of disadvantage - the variable around which the Brighton and Hove policy was built - is a distant contributor whose sign (direction of effect), as we discuss next, actually depends on the pupil group. Recognising this ordering matters, because a policy debate anchored on disadvantage concentration is, on this evidence, anchored on one of the weakest of the levers available.

The most contentious finding in our analysis concerns concentrations of disadvantage (contentious in Brighton and Hove - although the findings echo the analysis of Macleod et al. (2015) a decade ago). For non-disadvantaged pupils, higher FSM proportions are associated with lower attainment. But for disadvantaged pupils, the coefficient is *positive* - after controlling for absence and mean KS2 prior attainment, disadvantaged pupils perform slightly better in schools with higher concentrations of other disadvantaged pupils. This finding is robust across individual year models and alternative specifications of disadvantaged attainment and disadvantage measures.[11] In the full multilevel model, this positive coefficient shrinks substantially, suggesting the effect operates through LEA-level and Ofsted-level factors rather than concentration *per se* - schools with higher FSM proportions in certain LEAs appear to have developed effective support systems, possibly through targeted use of Pupil Premium funding and other specialisations.

We should be careful about the interpretation of this positive coefficient. One reading is that equivalent disadvantaged pupils do genuinely better in higher-concentration schools ('differing effects'). But an alternative is that the FSM binary cannot distinguish the depth or duration of disadvantage, so the disadvantaged pupils in a 50%-FSM school may differ systematically from those in a 20%-FSM school in ways our aggregate measures do not capture ('differing pupils'). With school-level data this is impossible to unpick definitively. However, the consistency of the result across multiple operationalisations of both disadvantage and disadvantaged attainment makes a pure measurement artefact less likely, and the stronger positive association we observe on the vocational/technical component of Attainment 8 (supplementary material) is at least suggestive of a genuine specialisation mechanism rather than pure composition. But the composition interpretation cannot be excluded, and distinguishing the two would require pupil-level data with disadvantage histories. Importantly, our policy argument does not depend on which interpretation holds: under either, there is no evidence that concentrating disadvantaged pupils *harms* their attainment, and therefore no support for the premise that de-concentration would *improve* it.

For non-disadvantaged pupils, the effect of higher disadvantage concentrations is more clearly negative - being in a higher-disadvantage school depresses Attainment 8 for this group, possibly through peer effects or curriculum-resourcing dynamics. Both findings challenge the 'no cost' conjecture and, in the Brighton and Hove context, suggest that further mixing of an already comparatively well-integrated set of schools could be counter-productive for disadvantaged attainment.

Absence is the biggest modifiable predictor across all groups and is more important still for

[11] See supplementary material for robustness checks: https://osf.io/x82j3/overview (see 'model_experiments.html' in 'supplementary.zip')

disadvantaged pupils, for whom it accounts for roughly a third of the combined standardised effect of every predictor in the model. Disadvantaged pupils may lack the safety net of engaged parents, private tutors or revision resources that some peers have, making classroom instruction more critical. Attendance, in this sense, is the ultimate equity lever. We do not advocate concentrating disadvantaged pupils, but the evidence does not support deconcentration policies premised on the assumption that disadvantaged pupils inevitably fare worse in higher-disadvantage schools. Where such policies require longer journeys that risk increasing absence (Thomson 2023), they could be actively counter-productive.

5 Brighton and Hove case study

5.1 Local context

Brighton and Hove has ten secondary schools: six community schools run by the LEA, two academies (Brighton Aldridge Community Academy - BACA - and Portslade Aldridge Community Academy - PACA) and two church schools (King's and Cardinal Newman) that set their own admissions. Following the closure of CoMArt in the east of the city in 2005, large catchment areas were introduced (Figure 2) with a lottery tie-break for oversubscribed schools - making admissions very different to most other places in England where distance typically determines priority (Allen et al. 2013).

In October 2024, the council launched proposals to amend the secondary school admissions process. The cabinet papers[12] set out three stated objectives: *'a school system where all pupils get access to a great education'*, improving *'the education offer for disadvantaged pupils by reducing some schools' barriers to success'*, and *'better equality of outcomes - results not driven by economic advantage'*. Alongside ran a School Organisation Strategy commitment to *'schools which are sustainable and able to thrive'* against falling pupil numbers. The specific proposals were to reduce Published Admission Numbers (PANs) at Longhill, Blatchington Mill and Dorothy Stringer by a combined 120 places, redraw the Longhill/Dorothy Stringer/Varndean boundary, and reserve up to 20% of places in single-school catchment areas for out-of-catchment children. Underpinning the case was a reading of Gorard and Siddiqui (2019) in which more mixed intakes produce better outcomes for disadvantaged pupils, plus a premise that attainment in the city was *'driven by economic advantage'* - a framing that implied that those who could afford to do so paid a housing premium to be near better schools in the city, but a characterisation empirically challenged in this context by Tan and Dennett (2025) who found that proximity to good schools in the city had a very small impact on prices. The 2024 proposals built on a 2023 policy giving FSM-eligible pupils priority at schools below the city-average (30%) FSM share, which was already producing flows from peripheral to central catchments.

The public response was overwhelmingly negative, with the council's pre-consultation feedback

[12] Brighton and Hove City Council (2024) *School Admission Arrangements 2026/27* - Cabinet Report, 5 December 2024 (decision to consult): https://democracy.brighton-hove.gov.uk/documents/s204040/School%20Admission%20Arrangements%202026-27.pdf. Brighton and Hove City Council (2025) *School Admission Arrangements 2026/27* - Full Council Report, 27 February 2025 (determination): https://democracy.brighton-hove.gov.uk/documents/s205834/School%20Admission%20Arrangements%202026-27.pdf.

summary noting *'a strong preference for improving existing schools rather than redistributing students, alongside deep concerns about potential impacts on community cohesion and student wellbeing'*. The proposals would, by the council's own calculations, force significant numbers of children from central catchments to schools at the city's periphery, potentially requiring over two hours of daily commuting. The council's transport-implications appendix[13] reports approximately £606,000 already spent each year on supported bus routes and bus passes, with further travel-assistance spending likely to support new pupil flows generated by the admissions changes. Notably, the previous administration's comprehensive strategy for tackling educational disadvantage[14] - which included sections on attendance, leadership, governance and targeted support - was not referenced in the 2024 consultation materials.

Whether better evidence would have changed the political outcome is unknowable, but accessible evidence would have enabled earlier and more constructive engagement. The compressed consultation timeline left inadequate time to assemble and present counter-evidence for the complexity of the issues at stake. What follows demonstrates what that evidence looks like and how it could inform future debates in Brighton and Hove and beyond.

[Figure 2: Brighton and Hove secondary schools with 2025/26 catchment boundaries. — about here]

5.2 Local authority effects

[Figure 3: LA random intercepts for disadvantaged-pupil attainment (imputed full panel model). Top and bottom 10 LAs labelled; Brighton and Hove highlighted. — about here]

Figure 3 shows the local-authority random intercepts for disadvantaged-pupil attainment. Conditional on the structural factors in the model - including school absence - Brighton and Hove ranks 15th of 151 local authorities (top 10%), a positive random intercept equivalent to roughly +1.6 Attainment 8 points above what those factors would predict. On the same conditional basis the city ranks 4th for non-disadvantaged pupils and 5th for all pupils. This was unknown at the time of the 2024 consultation and entirely absent from a public narrative framed around the raw attainment gap alone.

That conditional ranking, however, depends on a modelling choice that turns out to carry most of the result. Controlling for absence estimates the city's performance *given* the attendance its pupils actually record - and Brighton and Hove has among the worst attendance of any authority in England. Table 2 reports the disadvantaged-attainment effect with and without that control. Without any absence term, the city falls from 15th to 69th of 151 and its effect is no longer statistically distinguishable from the national average. The city's apparent excellence for disadvantaged pupils is, in this precise sense, conditional on setting its attendance problem aside.

[13] Brighton and Hove City Council (2024) *School Admission Arrangements 2026/27 - Appendix 9: Transport Implications and Considerations*. Cabinet papers accompanying the secondary school admissions consultation, available via the consultation portal: https://yourvoice.brighton-hove.gov.uk/en-GB/projects/school-admission-arrangements-public-consultation/1.

[14] https://www.brighton-hove.gov.uk/schools-and-learning/school-policies-reports-strategies-and-other-documents/better-outcomes-better-lives-brighton-hoves-strategy-tackling-educational-disadvantage

[Table 2: Brighton and Hove's local-authority effect for disadvantaged-pupil attainment, with and without a control for school absence. The intake-predicted specification controls only for the part of absence that intake predicts, leaving the city's excess absence in the residual. — about here]

This is not an artefact peculiar to Brighton and Hove: controlling for absence reshuffles the whole distribution of local-authority effects, and 8 authorities that appear significantly above average once absence is controlled are indistinguishable from average without it. Absence does a great deal of work in any model of attainment, and where an authority sits on it materially changes how its schools appear to perform. But the shift is unusually consequential for Brighton and Hove precisely because its attendance is so extreme.

The distance between these two numbers is the important finding, rather than either number alone. Once we account for the absence its pupils experience, Brighton and Hove's schools sit within the strongest 10% of local education authorities in England for disadvantaged attainment; if we don't account for this, they are unremarkable. The gap between the two - about 1.4 Attainment 8 points per disadvantaged pupil - is a direct measure of what the city's attendance problem costs its most disadvantaged children. It is not a reason to discount the schools' underlying effectiveness; it is a reason to treat attendance as the single most valuable lever available to the city, a point we develop in section 5.5 and quantify through the accelerating-returns analysis below. (For all pupils the city's advantage is more robust to this choice, remaining significantly positive without an absence control at 69th; it is specifically the disadvantaged-pupil result that rests on it.)

Two features of the city's absence confirm that this is a local problem rather than an inevitability of its intake. First, Brighton and Hove's disadvantage is unexceptional: its mean school-level free-school-meal rate (25.2%) is slightly *below* the national average (27.1%). When we model expected absence from intake alone (section 5.5), the city's schools should record about 8.8% absence; they actually record 10.8%. Almost two percentage points of absence sit above what the intake predicts. Second, the problem has not been improving. Between 2021–22 and 2024–25, absence fell in 94% of English local authorities - the national mean dropping by 0.88 percentage points as attendance recovered from the pandemic - while Brighton and Hove's barely moved (-0.03 points), leaving it 143th of 152 authorities for improvement. The city did not become an outlier because its absence rose; it became an outlier because, almost alone, its absence did not fall.

5.3 Comparing policy levers: absence versus disadvantage

Because the model uses log-transformed variables, the same percentage-point change produces different effects depending on where a school starts - returns *accelerate* as values improve. Figure 4 places both the absence and FSM levers on a common y-axis.

[Figure 4: Accelerating returns: each percentage-point reduction buys a progressively larger ATT8 gain (reading right to left). Absence delivers substantially larger gains than FSM at every level. Brighton and Hove schools marked on each curve. — about here]

Four features are immediately visible. First, the absence curve sits above the FSM curve at every starting level, indicating how much more effective changes in absence are than changes in disadvantage, for improving attainment. Second, the gap *widens* as values improve. Third, schools with the highest proportions of disadvantage and to a lesser extent, absence, will notice the least impact on attainment if those levels change. Fourth, Brighton and Hove schools cluster in the flatter right-hand portion for both variables, but for every school, changes in absence are likely to affect more noticable changes in attainment than changes in disadvantage.

Brighton and Hove's mean school FSM rate (28.8%) is close to the national average (28.3%). But the city's mean absence rate (10.8%) against a national average of 8.2% places it at the 99th percentile - among the very worst in England. When standardised by the city's spread, the absence coefficient is 3.4 times as important as FSM for predicting attainment.

Bringing schools above the national average for absence down to it would produce predicted gains averaging +2.8 Attainment 8 points per school, with gains for disadvantaged pupils averaging +3.3 points. Even dramatic FSM reductions would yield gains closer to 1 point. On the evidence, attendance is the single most consequential lever the city has available, and admissions changes of the kind proposed in 2024 are very unlikely to produce equivalent gains in disadvantaged attainment.

5.4 Contextualised school performance

[Figure 5: Observed versus predicted Attainment 8 for Brighton and Hove schools (highlighted) against all schools nationally, 2024–25. — about here]

Figure 5 shows observed versus predicted Attainment 8 for the latest year across all schools in England (grey) with Brighton and Hove schools highlighted (pink). The close clustering of the points around the diagonal line is a visual representation of the R^2 values show on the plot and in the tabular model outputs shown earlier - observed values for most schools are close to the model predictions. The model residuals - the vertical distance from the diagonal - show whether a school is doing better (above) or worse (below) than the model predicts and provides what we might think of as contextualised 'value-added' measures - more nuanced than Progress 8 - as this model-based value added accounts for far more than just prior attainment in its representation.

5.5 Decomposing absence: structural intake versus school management

The contextualised value-added reading above hides a complication. Our headline model treats school-level absence as a single factor on the same footing as other intake variables, but absence is different to FSM or EAL. Part of it is exogenous to the school - driven by family circumstances, area-level health, deprivation, and SEND profile - while part is produced by the school via pastoral systems, attendance officers, parental engagement and the broader ethos around getting children into class. Controlling for raw absence absorbs both components together, so the residual we have called 'value-added' reflects what the school does *given* its attendance, not what it does *to* its attendance.

Two parallel models[15] separate these elements from each other and in doing so we adjust for what a school inherits, not for what it does. Firstly, we model school-level absence on the variables we treat as exogenous (FSM, EAL, mean KS2 prior attainment, segregation, plus place and year random effects), producing for every school-year an *expected absence* (intake-predicted - around 55% of the variation) and a *residual/excess absence* — the part beyond what its intake would predict, which is the school's own contribution to its attendance (around 45% of the variation). Secondly, we fit the attainment model on the same intake variables but with *no* absence control and no workforce predictors, since attendance management and workforce decisions are things a school does rather than things it receives. The school-level residual from this second model is a more honest value-added measure: it is the school's total contribution to attainment, with attendance management counted inside that contribution rather than stripped away as a control.

There are two issues with this that we should make clear. The first is because neither sub-model controls for observed absence, any absence that is genuinely outside a school's control, but not captured by intake or place (for example SEND related absence), will be charged to the school's value added. The first-stage model explains 55% of absence variation, so what remains unexplained is quite large and thus could give a false impression of value added. The second issue relates to how we interpret the coefficients in this exercise - they can't simply be interpreted as effects. We know how important absence is from our earlier models, so leaving it out doesn't remove its influence, that influence is redistributed across the variables that remain. And because disadvantage is correlated with absence, that then absorbs most of the effect Therefore we only take the residuals from this model. Every disadvantage estimate reported elsewhere in this paper comes from the headline specification, which controls absence directly.

The national picture is informative. Most of the school-year variation in absence is explained by the intake-only first stage; the school-controllable residual is the smaller share, and its marginal contribution to attainment is modest once intake-driven absence is accounted for. The headline raw-absence coefficient therefore borrows much of its statistical force from the structural component: the lever is real, but the school-only portion is smaller than the single coefficient implies. For LEAs this means closing a city's absence problem requires both school-level attention *and* cross-departmental work - public health, children's services, area-level deprivation policy - because much of what schools deal with on attendance is inherited rather than generated.

[Figure 6: Brighton and Hove secondaries on the joint-signal plane: contextualised value-added (vertical, in Attainment 8 points) versus the school-controllable absence component (horizontal, in percentage points). The reference lines at zero on each axis split the plane into four narrative quadrants discussed in the text. The full national version with a local-authority filter is available interactively in the project's HTML report. — about here]

For local school-level diagnostics the decomposition becomes considerably more informative. Two signals can now be read side by side: the school's contextualised value-added, and its

[15] Full technical details, model fits, sensitivity checks and a 50-replicate clustered bootstrap of school rank uncertainty are in the supplementary material: https://osf.io/x82j3/overview (see 'model_experiments.html#sec-two-stage-absence' in 'supplementary.zip')

residual absence (positive when the school has more absence than its intake would predict, negative when less). Sorted on these two axes (Figure 6), schools fall into four meaningful policy stories: schools adding value with attendance management helping (Q1, top-left); schools adding value despite worse-than-predicted attendance (Q2, top-right); schools underperforming with attendance the cleanest single lever (Q3, bottom-right); and schools underperforming despite unusually good attendance management (Q4, bottom-left).

For Brighton and Hove the joint-signal view sharpens the case-study reading. Most of the city's schools sit in Q2 in the top-right: they are delivering attainment above what their intake would predict and are doing so while experiencing more absence than their intake-driven absence model expects. In these schools, the pedagogical work happening is the more impressive for it; closing any of these school's residual-absence gap would compound directly onto an already-strong value-added signal. Any schools in Q4 provide a challenge: their attendance management is genuinely strong, but once intake is properly accounted for their attainment outcomes fall short of what their profile predicts, so the simple "fix attendance" improvement story is not available. Schools in Q3 are the most natural targets for council-led attendance interventions, and Q1 schools are where the city should look for practice worth sharing.

We are aware that Progress 8 is an attempt to bring a degree of value added into the reporting of attainment, but we would advocate for a more sophisticated, model-based measure of value added as outlined here. Controlling for many more of the exogenous variables that influence attainment allow us to dig deeper into the impact that schools are having, and as we have shown here, the reporting of measures such as value-added alongside residual absence means we can achieve a more honest framing than the single number that conventional benchmarking produces. This would help align local policy conversations with what the data can support them.

6 Discussion

6.1 The danger of single-lever policy thinking

The Brighton and Hove experience illustrates the risks of building a policy around a single causal narrative. The council's 2024 admissions proposals were built around the premise that redistributing disadvantaged pupils would narrow the attainment gap. In the consultation papers released to the public[16], it was stated:

> *"We also know that children from disadvantaged homes are less likely to do well at school and that can be made worse when more disadvantaged children all go to school together. We are looking at options for how we can help the city manage that issue better, including through our school admissions arrangements."*

This premise sits awkwardly with the existing evidence base. Macleod et al. (2015)'s earlier analysis for the DfE found that, controlling for intake characteristics, schools with higher

[16] Brighton and Hove Council 2024 Consultation Papers: https://yourvoice.brighton-hove.gov.uk/en-GB/projects/school-admission-arrangements-public-consultation/1

concentrations of disadvantage tend to do *better* for their disadvantaged pupils, not worse. Our more up-to-date modelling on four years of national data reaches the same conclusion. When it comes to attainment, concentrations of disadvantage are one of the weaker available levers, and for disadvantaged pupils specifically the direction of the effect runs in the opposite direction to the policy assumption. The implication is that admissions changes of the kind proposed in Brighton and Hove are unlikely to deliver the targeted gains in disadvantaged attainment and could produce unintended aversive outcomes.

It is worth setting this in the longer context of what school-effectiveness research has consistently found, because our result is not a novel or contrarian one. Ever since Coleman et al. (1966), the share of variation in pupil outcomes attributable to differences *between* schools - as opposed to differences in the pupils they admit and the circumstances those pupils bring - has been estimated at something in the order of 8 to 15 per cent. The landmark English study by Smith and Tomlinson (1989) settled on around a tenth, and Teddlie and Reynolds (2000) confirm that broad range across countries and phases. Our own between-school decomposition using recent data in England, reported in the supplementary material, arrives at something similar. This matters for how much any redistribution policy based on school performance measures can be expected to achieve: if differences in pupil outcomes between schools are little to do with what those schools do but are instead driven by wider societal impacts on the students they are admitting, then a policy that redistributes pupils between schools operates on a comparatively small share of the variance in the first place - it is working on the smaller term while leaving the harder, larger problem untouched. It will be an ineffective plaster stuck on the wider problem of social inequalities.

Furthermore, the chosen lever also carries risks that the proposals did not engage with. Redistribution to schools outside of a pupil's local area generates longer journeys to school, and the wider literature is clear that long commutes are associated with reduced sleep and its psychosocial (Fredriksen et al. 2004; Yeo et al. 2019), physical (Voulgaris et al. 2019; Pradhan and Sinha 2017; Pereira et al. 2014; Faulkner et al. 2013) and mental health (Nakajima et al. 2024; Chairassamee et al. 2024; Guan et al. 2025) consequences, including stress (Karan et al. 2021), lower wellbeing (Ding et al. 2023), and higher absence across multiple national settings (Otsuka et al. 2025; Cordes et al. 2022; Blagg et al. 2018), including England (Thomson 2023). Absence is the largest predictor of attainment in our modelling and the city's most acute structural problem; a policy that even marginally increases it could be counterproductive. There is also a qualitative dimension: schools serve as community anchors, supporting after-school participation, friendship groups, parental engagement and active travel (Faulkner et al. 2013). A Chicago case study (Shah 2023) argues that the burden of long commutes falls disproportionately on pupils already experiencing other forms of marginalisation, and long-distance transit predicts early high school transfer (Stein et al. 2021), itself a risk factor for poor attainment. These are not peripheral concerns; they, along with another other potential benefits, are part of the cost-benefit calculus any redistribution policy needs to engage with, but were largely missing from the 2024 deliberation in Brighton and Hove. The claim that social mixing can be achieved at 'almost no cost' (Gorard and Siddiqui 2019) sits awkwardly with this second-order literature.

It is worth making the mechanism explicit, because it clarifies how an admissions reform actually reaches attainment. There are two pathways. The first is the intended one: changing where pupils go alters the *FSM composition* of individual schools, which our modelling shows has, at most, a small effect on attainment - and for disadvantaged pupils, an effect that if anything runs the wrong way. The second is unintended: moving pupils further from home increases *travel*, which the evidence above links to *higher absence.* Recent research also clearly links attendance with the sense of well-being, life satisfaction and belonging in students (Moore et al. 2026). While it may not necessarily follow that this will be diminished in students attending schools far from their home communities and established friendship groups, it is reasonable to hypothesise this as a further risk factor to consider, particularly where absence is by a wide margin the strongest predictor of attainment we identify. In other words, the data suggests that the admissions lever operates on attainment mainly through the second, unintended channel, and in the direction opposite to the policy's aim. This is why we describe it as pulling the wrong lever: not merely a weak instrument, but one whose most probable net effect on disadvantaged attainment is negative. We should be clear that this is a statement about *expected* effects derived from national associations, not an evaluation of the 2023 free-school-meal-priority policy or the 2026 arrangements, which post-date our outcome data and whose effects the council's own commissioned follow-up work has yet to report.

6.2 Attendance as the most impactful single lever

A policy conversation about Brighton and Hove's disadvantage attainment gap that wants to make the largest predictable difference should put attendance at its centre. This sits squarely within the council's own framing: the December 2024 cabinet report named *'reducing some schools' barriers to success'* for disadvantaged pupils as one of its three core objectives for the 2026/27 admissions changes [17]. Of the school-level barriers our modelling is able to quantify - intake composition, prior attainment, workforce stability, segregation, and absence - none has a larger or better-evidenced effect on disadvantaged attainment than absence. On the council's own framing, in other words, the most direct route to *'reducing barriers to success'* for the city's disadvantaged pupils points at attendance rather than at admissions arrangements. The city has the second-worst absence rate in England in 2024-25 - and, unlike almost every other authority, one that has barely improved since the pandemic - and the wider evidence indicates that closing this gap would yield attainment gains for disadvantaged pupils that comfortably exceed anything achievable through changes to admissions arrangements - and would do so *without* the social, health and travel costs that a redistribution-led approach might produce.

We should reiterate that Brighton and Hove's absence rates are not a necessary consequence of its intake. One response to an earlier presentation of this argument, advanced locally by the campaign group Class Divide[18], offered a memorable analogy: absence is the mould, poverty is the damp. On this view, treating absence directly is futile, because the underlying

[17] Brighton and Hove City Council (2024) *School Admission Arrangements 2026/27* - Cabinet Report, 5 December 2024 (decision to consult): https://democracy.brighton-hove.gov.uk/documents/s204040/School%20Admission%20Arrangements%202026-27.pdf. Brighton and Hove City Council (2025) *School Admission Arrangements 2026/27* - Full Council Report, 27 February 2025 (determination): https://democracy.brighton-hove.gov.uk/documents/s205834/School%20Admission%20Arrangements%202026-27.pdf.

[18] The comment was made in a blog post here: https://www.classdivide.co.uk/news/uk5uk5bxax0rvjz9q4tu4kgx2lbyhw

poverty will simply keep bringing it back. The analogy captures something real. At pupil level the association between disadvantage and absence is strong and well evidenced (DfE 2025b), and our own decomposition (section 5.5) finds that a substantial part of school-level absence is inherited from intake rather than generated by the school. Any serious attendance strategy must therefore engage with the material circumstances of families, not merely with attendance procedures.

It is important, however, to consider that effect of absence on attainment in our model is the effect above and beyond being eligible free school meals, the best measure we have of poverty in our school intakes. Some variability of absence and poverty in the intake are shared. Absence may be a symptom of poverty, but it is also a likely mechanism of poverty's effects on attainment. As such, absence is also a target for intervention. Any serious attendance strategy must engage with the material circumstances of families, not merely with attendance procedures. However, beyond addressing the hard problem of economic disadvantage that can underpin many factors in the attainment model, increasing attendance of students may go the furthest to address the disadvantage attainment gap.

But as clever as the analogy is in general terms, it does not explain the specific situation in Brighton and Hove. Across English local education authorities, differences in poverty account for remarkably little of the difference in absence: regressing LEA mean absence on LEA mean FSM eligibility across the 152 authorities in our latest year produces an R^2 of just 0.05. Brighton and Hove's own position makes the point clearly. Its FSM rate (28.8%) is slightly *below* the average across authorities (29.6%), so on a poverty-only account its absence should be a little below average too - around 8.2%. It is 10.8%, an excess of 2.6 percentage points over what its disadvantage profile predicts, and the largest such excess of any authority in England. Among the 20 authorities whose FSM rates are within two percentage points of Brighton and Hove's, the city has the *highest* absence of all, some 2.3 percentage points above their average. And where absence fell in 94% of local authorities between 2021–22 and 2024–25, the city's did not move. At the risk of stretching the metaphor too far, there is damp in many places but the mould is markedly worse here, and it is not getting better as it is elsewhere.

The reasonable conclusion is not that poverty is irrelevant to absence, but that it cannot be the full explanation for Brighton and Hove's position. Treating the city's absence problem as an inevitable consequence of its social composition is empirically unsupported and risks diverting attention and resources away from what should be the laser focus of the LEA. A substantial part of the city's absence sits above what its intake predicts, which is precisely the part that local action can reach. The problem therefore reflects local factors - attendance culture, enforcement practices, transport friction or community dynamics - rather than being an unavoidable outcome of disadvantage. If the city is not already running a substantial cross-departmental attendance programme alongside its school-level work, the case for one is strong to introduce one. There is a budget question too: the £606,000 already spent annually on supported bus routes and passes[19], plus any further travel-assistance spending

[19] Brighton and Hove City Council (2024) *School Admission Arrangements 2026/27 - Appendix 9: Transport Implications and Considerations.* Cabinet papers accompanying the secondary school admissions consultation, available via the consultation portal: https://yourvoice.brighton-hove.gov.uk/en-GB/projects/school-

generated by the admissions changes, would arguably have a stronger evidence base for raising disadvantaged attainment if invested directly in attendance support, family liaison and pastoral capacity in the city's most affected schools.

6.3 A new value-added view to support better local decisions

Another contribution of this work is methodological. The multilevel model lets us look beyond a school's raw attainment outcomes and ask how much it is adding *over and above* the structural intake characteristics it inherits - characteristics that the school cannot itself choose. The school-level random intercept, plus the school-level residual from the fitted model, gives us a contextualised value-added view that is fairer than the headline attainment number and more informative than a single (or multi)-word Ofsted rating, both for parents making school choices and for policy makers assessing where local effort is best directed. Brighton and Hove provides a particularly sharp illustration of why this matters: the local school for one of the city's most disadvantaged catchments is, on this view, doing materially better for its disadvantaged pupils than the schools to which a number of those pupils' families have been opting to travel.

[Table 3: Value-added rankings for Brighton and Hove schools (panel model residuals by year, in Attainment 8 points). — about here]

During the 2024 consultation, the public debate shifted from how the council's proposals would improve disadvantaged attainment to how they would facilitate parental choice. An activist group (Equity in Education[20] - supported by Class Divide) emerged from within the BACA catchment advocating for alternatives to that school. While each parent has their reasons for wanting a different choice, at the time, BACA had a 'requires improvement' Ofsted rating and some of the city's lowest raw attainment scores. A considerable number of BACA-catchment parents subsequently opted for Patcham School - 'Good' rated, mid-table on raw attainment, in a more prosperous neighbouring catchment.

However, the contextualised picture differs sharply. BACA sits in one of the most deprived parts of the city, experiences absence well above the high city average, has around 50% FSM and 30% low prior attainment; Yet, BACA out-performs every other school in the city for disadvantaged pupils, including those popularly viewed as the 'best'. Once these cohort differences are accounted for, the league tables in Table 3 (derived from Figure 5) show that in 2024-25 a disadvantaged pupil at BACA left, on average, with GCSE results around 5.7 points higher than a comparable pupil at a similar school elsewhere in England, with a four-year average around 3.1 points higher - substantially above the fraction of a point that might be expected from altering schools' disadvantage mix. Its April 2025 Ofsted upgrade to 'Good' in all five areas[21] is unsurprising in the context of this analysis, but would have been surprising to anyone looking only at prior raw attainment and Ofsted ratings.

admission-arrangements-public-consultation/1.

[20] https://educationequity.kit.com/

[21] https://reports.ofsted.gov.uk/provider/23/136164

6.4 Viability versus attainment

The Brighton and Hove City Council cabinet papers of 5th December 2024[22] set out the financial-sustainability rationale plainly. Schools in England are funded almost entirely on a per-pupil basis, so falling pupil numbers across the city would, in time, leave some schools struggling to operate efficiently or *'offer a wide and balanced curriculum'*. The proposed PAN reductions of 120 places in total were framed by the council as a way of distributing that impact across the city's most over-supplied schools rather than concentrating it on individual under-subscribed ones. The council also noted, importantly for the framing of the policy, that *'the largest percentage of [disadvantaged] pupils attend the schools with the lowest pupil numbers'* — making the viability concern and the disadvantage objective explicitly interlinked in the council's argument. Financial viability is therefore a legitimate concern, particularly given the historical context of an earlier school closure in a deprived part of the city that left some pupils already travelling long distances to peripheral provision. There is, however, an inherent tension between optimising for financial sustainability and optimising for educational outcomes — and the two cases need to be weighed transparently. The contextualised value-added view introduced above suggests that the city's pattern of school-level effectiveness does not map cleanly onto raw attainment or popularity: if redistribution moves children from schools adding more value to schools adding less, the aggregate impact on city-wide attainment could be negative even if the financial case for the receiving schools improves.

Following widespread opposition and appeals to the Schools Adjudicator, the proposed PAN reductions were rejected. Parental objections meant that the 20% out-of-catchment priority was reduced to 5%. While displacement was minimised, the policy reverberations continued through choices expressed by parents already influenced by the preceding debate.

6.5 Conditional performance and the cost of absence

The LEA random effect for Brighton and Hove is positive and, in the main model, statistically significant across all three attainment groups. As section 5.2 set out, however, this result is conditional on how the model treats absence, and the city's attendance is so extreme that the conditioning does most of the work. Controlling for absence, the city's schools rank 15th of 151 for how far disadvantaged attainment exceeds what intake predicts; without that control they rank 69th and are statistically ordinary. We take neither number as the whole truth. Together they say something more useful than either alone: Brighton and Hove's schools appear to be genuinely effective for the pupils who attend them, and the city's attendance problem is large enough to erase that advantage in the aggregate.

The distance between the two rankings has a concrete interpretation. It is roughly 1.4 Attainment 8 points per disadvantaged pupil - the difference between what the city's disadvantaged children achieve and what they might achieve if their schools' underlying effectiveness were not being drained by absence. Read this way, the city's attendance is not a caveat to a good-news story; it is the story.

[22] Brighton and Hove City Council (2024) *School Admission Arrangements 2026/27* - Cabinet Report, 5 December 2024 (decision to consult): https://democracy.brighton-hove.gov.uk/documents/s204040/School%20Admission%20Arrangements%202026-27.pdf

This reframes what the 2024 consultation got right and what it got wrong. The council was not wrong that Brighton and Hove has a problem: unconditionally (without absence in particular), its disadvantaged pupils do no better than the national average, and the raw attainment gap that motivated the consultation is real. What the consultation got wrong was the diagnosis. Presented with an attainment gap, it reached for the social composition of schools at considerable cost - one of the weakest levers our modelling identifies, and for disadvantaged pupils a lever that may point the wrong way (section 4.3, section 5.3). Meanwhile the factor that actually separates the city's conditional standing from its unconditional standing, attendance, was absent from the consultation entirely. The more accurate account is not that a high-performing city was mistaken for a failing one, but that a city with one specific, severe, addressable problem was prescribed a treatment for a different problem it did not have.

The honest answer is that we do not yet know what produces differences in school attainment beyond our model. The random effect, by definition, captures variance that the fixed-effect predictors do not explain. Good performance could reflect high levels of parental engagement and aspiration and a strong civic culture. It could relate to the quality of school leadership and governance across the city, or to effective local authority support services and school improvement programmes. It could be something about the collaborative relationships between schools, or community factors that are difficult to quantify. It could even be that if otherwise disruptive students switch to being absent students at a higher rate in Brighton and Hove, this may even have a positive impact on learning for those who remain in school. Without pupil and class-level longitudinal analysis, any hypothesis would be impossible to test. Identifying the sources of conditional over-performance is an important research question in its own right, and one that could require additional qualitative investigation or access to pupil-level data that goes beyond what is publicly available. But what is clear is that policy interventions which risk disrupting a system without understanding what makes it function well carry substantial downside risks.

6.6 Implications for national policy

The DfE (2026) white paper's ambitions will founder if pursued at the local level without adequate analytical infrastructure. Every LEA occupies different positions along the non-linear curves that relate predictors to outcomes. What works in one context may be counterproductive in another. The DfE's open data provides most of the raw materials, but there is a crucial gap between raw data availability and the accessible, contextualised intelligence that decision makers need.

Academisation and sustained real-terms reductions in local authority central-services funding is in part responsible for this gap, having substantially eroded the analytical capacity that once sat within LEAs. Council's frequently remain the admissions authority for systems that funding cuts and academisation mean they have limited capacity to analyse as a whole. As such a recommendation simply to 'invest locally' sits uneasily with the fiscal reality most authorities face. Multi-academy trusts have absorbed some of this function, but only for their own constituent schools; no actor now has both the remit and the capacity for the *locality-wide* analysis that spans maintained schools, academies across different trusts, and church schools within a single place. Brighton and Hove illustrates the gap to some extent – although where it still has direct control of 6 out of 10 community secondary schools and academisation has not (yet) had the same impact as it has elsewhere, to a lesser degree than many.

But LEAs can also do much more to help themselves where capacity and funding constraints are real. While we have pieced together a national analysis here from DfE data and applied it to a local context, Local Authorities could put more local data accessibly into the public domain. Freedom of Information Requests should not be the default route for those in the wider community - potentially with valuable time and expertise - to access important local information. Data such as aggregate pupil projections for schools and neighbourhoods, time series preference data, distances travelled to school, offers by school and home neighbourhood deprivation band, Free School Meal eligibility by neighbourhood, appeals and their outcomes – these are all held by LEAs but rarely published as a matter of course. Hiding data away either completely or in inaccessible PDFs tucked away in committee meeting appendices, rather than publishing openly in machine-readable formats is counter-productive. In an effort to demonstrate how our collective intelligence could be enhanced through better data analytics we have developed an interactive Policy Simulator tool,[23] underpinned by the data and models in this paper. The tool allows users to explore any school in England, compare against contextual expectations, and model the indicative impact of changes to key variables. This is only a starting point and more bespoke intelligence systems could be built for local situations.

6.7 Limitations

These are school-level models built from aggregate published data, not pupil-level analyses. The associations identified are not experimental causal estimates but the best approximations available from observational data. The models cannot capture unmeasured factors such as school culture, parental engagement, or individual pupil resilience. The Policy Simulator translates associations into indicative, not definitive, projections. We do not claim that reducing absence by a given amount will mechanically produce the predicted gain - the real world is more complex than any model captures. What we do claim is that these models are accurate enough, and the patterns consistent enough across years and specifications, to substantially improve the quality of the conversation around what matters most.

We should also say something about statistical inference, given the nature of our data. Our panel is not a random sample drawn from some larger frame; it is a population, comprising every state-funded mainstream, non-selective secondary school in England. As such we do not interpret standard errors as quantifying uncertainty about generalising from a sample to a population. We interpret them as quantifying the stochastic, cohort-to-cohort variation that produces any given school's result in any given year: each cohort is one realisation of a process that would have turned out somewhat differently with a different set of children, and school-level scores are correspondingly unstable from year to year - as is observable from the variation across the four-year value added scores in Table 3. The multilevel structure of our model is an additional insurance, since the partial pooling inherent in these models

[23] School Attainment Policy Simulator: an interactive R Shiny web tool developed as part of this work. URL withheld for double-blind review; tool details, a walkthrough and access can be made available on request from the corresponding author via the editorial office.

shrinks the noisiest, most variable year-on-year estimates - typically those from the smallest schools - toward the relevant group mean. Conscious of long-standing cautions about the over-interpretation of p-values (Wasserstein and Lazar 2016), we treat significance in our model as descriptive rather than decisive. Readers will note that there is nothing in this paper turns on a p-value threshold being crossed. In addition, some may try to point out that coefficients are not effect-sizes, which is true, and why we have taken care to convert effects that are most relevant into more widely understandable Attainment 8 points. Our observations are reproduced across four separate years, and they would stand unchanged if every test statistic were removed.

7 Conclusion

This paper has demonstrated that the DfE's open data is a substantially underutilised resource for understanding school attainment and informing local policy, at least in the context of our case study LEA. School-level Attainment 8 is remarkably predictable from a small number of variables, and that predictability creates opportunities for better-informed local policy. The Brighton and Hove case study surfaces three observations we feel are well-supported by the evidence presented here.

First, the city's 2024 admissions proposals were built on a premise — that concentrations of disadvantage are a primary driver of low disadvantaged attainment — that does not hold in the context of four years of national data analysed here, nor with the earlier DfE-commissioned analysis (Macleod et al. 2015). Combined with the wider literature on the consequences of long school commutes for sleep, health, well-being, attendance, school transfer and the loss of community-anchored schooling, the proposals are unlikely to deliver the targeted gains in disadvantaged attainment and carry a set of social and educational costs that the consultation did not engage with.

Second, the single most consequential lever available to Brighton and Hove for raising disadvantaged attainment is attendance. The city had the second-worst absence rate in England in 2024-25 - and its absence has failed to fall while it fell almost everywhere else in England - and the modelling suggests that moving the city's worst-affected schools toward the national absence average would yield gains for disadvantaged pupils that comfortably exceed anything achievable through admissions reform. The absence decomposition we develop here also shows that this lever has two parts to it: a *structural* component of school-level absence inherited from intake, area health, deprivation and family circumstances - which schools cannot themselves change - and a smaller *school-controllable* residual that pastoral systems, attendance officers, family liaison and persistent-absentee follow-up can act on directly. Addressing the city's absence problem therefore needs both school-level effort and cross-council-department action; treating it as a single school-level lever would over-state what individual schools can deliver alone. If the city is not already running a substantial cross-departmental attendance programme, the case for one is strong; resources of the scale required to bus children between catchments would, on the evidence, deliver more for disadvantaged pupils if invested directly in attendance and pastoral support in the city's most affected schools.

Third, the modelling provides a contextualised, value-added view of school performance that is fairer than raw league tables and more informative than a single-word Ofsted rating. For parents this matters directly: in Brighton and Hove, the local school of one of the city's most disadvantaged catchments is, on this view, doing materially better for its disadvantaged pupils than the schools families have been opting to travel to instead. A wider understanding of this view would help families make decisions that work for their children *and* preserve the benefits of local schooling.

Our recommendations follow. For the DfE: invest in analytical infrastructure at the local level and consider how contextualised benchmarking tools might be developed at scale. For LEAs: contextualised benchmarking and a focus on the most impactful levers should be at the centre of local policy on attainment. The factors affecting attainment are multiple, interacting, non-linear and context-dependent, and a single-narrative approach risks pulling the wrong lever. For schools, governors and parents: a contextualised value-added view offers a fairer basis for understanding school quality than raw scores or Ofsted ratings, and can support better choices that do not come at the cost of community-anchored schooling, while helping point to where improvement efforts might be most usefully targeted.

Declaration of generative AI use

The authors used Anthropic's Claude Code (Claude Opus 4.6) to assist with the development of R data-processing pipelines, the implementation of multilevel model fitting and visualisation code, the formatting of tables and bibliography entries, and the rendering and deployment of Quarto outputs. Any analytical decisions and interpretations during the iterative analytical process were reviewed, verified and approved by the authors, who take full responsibility for the work.

Disclosure Statement

Some of the authors are residents of Brighton and Hove with school aged children and lived experience of the effects of the admissions policy process described. No financial conflicts of interest exist.

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Table 1: Stepwise progression to the full multilevel model for overall Attainment 8.

	M1: FSM		M2: +Absence		M3: +Prior		M4: All fixed		M5: Full LME	
Variable	Est.	t	Est.	t	Est.	t	Est.	t	Est.	t
(Intercept)	4.473 ***	621.25	5.006 ***	579.97	4.412 ***	345.92	4.304 ***	263.27	4.261 ***	114.98
log(% FSM)	-0.201 ***	-88.94	-0.122 ***	-59.66	-0.047 ***	-21.04	-0.054 ***	-21.47	-0.050 ***	-18.18
log(% Absence)			-0.364 ***	-82.87	-0.259 ***	-60.47	-0.229 ***	-49.98	-0.207 ***	-45.74
log(% EAL)							0.013 ***	12.83	0.007 ***	6.09
Mean KS2 Prior Attainment					0.031 ***	58.18	0.030 ***	46.70	0.032 ***	48.46
Admissions: Other non-selective							0.039 ***	8.73	0.028 ***	4.06
Admissions: Selective							0.019 *	2.44	-0.004	-0.55
Gorard Segregation							0.075 **	2.84	0.026	0.58
Teacher Retention							0.001 ***	11.66	0.000 ***	8.99
Leadership Pay %							-0.001 ***	-5.45	-0.001 ***	-4.52
log(Teacher Sickness)							-0.013 ***	-5.20	-0.013 ***	-5.69
Random effects										
Year									0.026	(SD)
Ofsted rating									0.014	(SD)
Region									0.045	(SD)
LA (nested in region)									0.045	(SD)
Residual									0.092	(SD)
R^2	0.393		0.612		0.696		0.709		0.651	(marginal)
R^2 (conditional)									0.779	(fixed + random)
N	12,210		12,210		12,210		12,210		12,210	

Figures and Tables

Table 2: Brighton and Hove's local-authority effect for disadvantaged-pupil attainment, with and without a control for school absence. The intake-predicted specification controls only for the part of absence that intake predicts, leaving the city's excess absence in the residual.

Specification	B&H effect	95% CI	Rank	ATT8 pts
No absence control	+0.006	-0.034 to 0.046	69 of 151 (n.s.)	+0.2
Absence controlled (main model)	+0.041	0.006 to 0.076	15 of 151	+1.6
Intake-predicted absence controlled	+0.005	-0.035 to 0.044	71 of 151 (n.s.)	+0.2

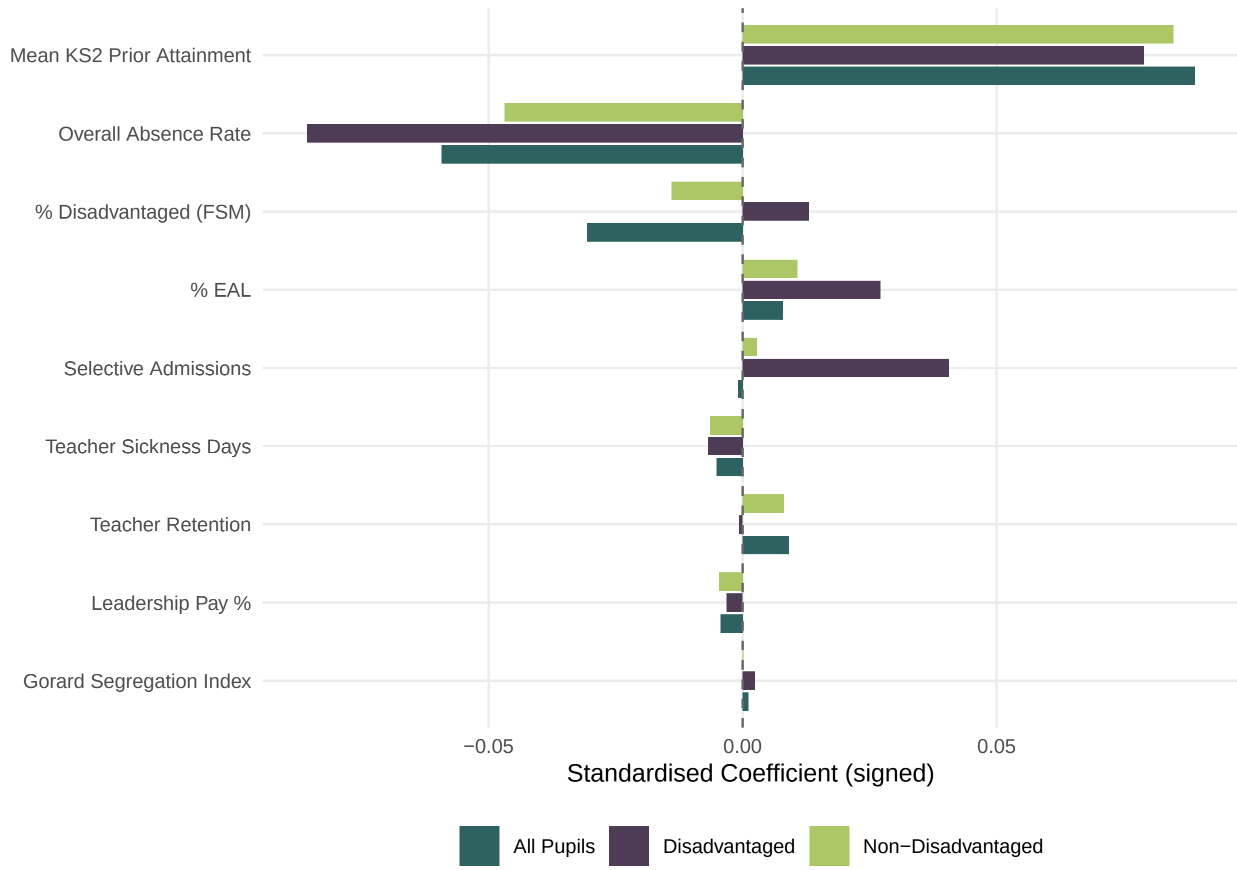

Figure 1: Standardised coefficients showing relative variable importance across pupil groups. Bars show the change in log(ATT8) associated with a one-SD shift in each predictor.

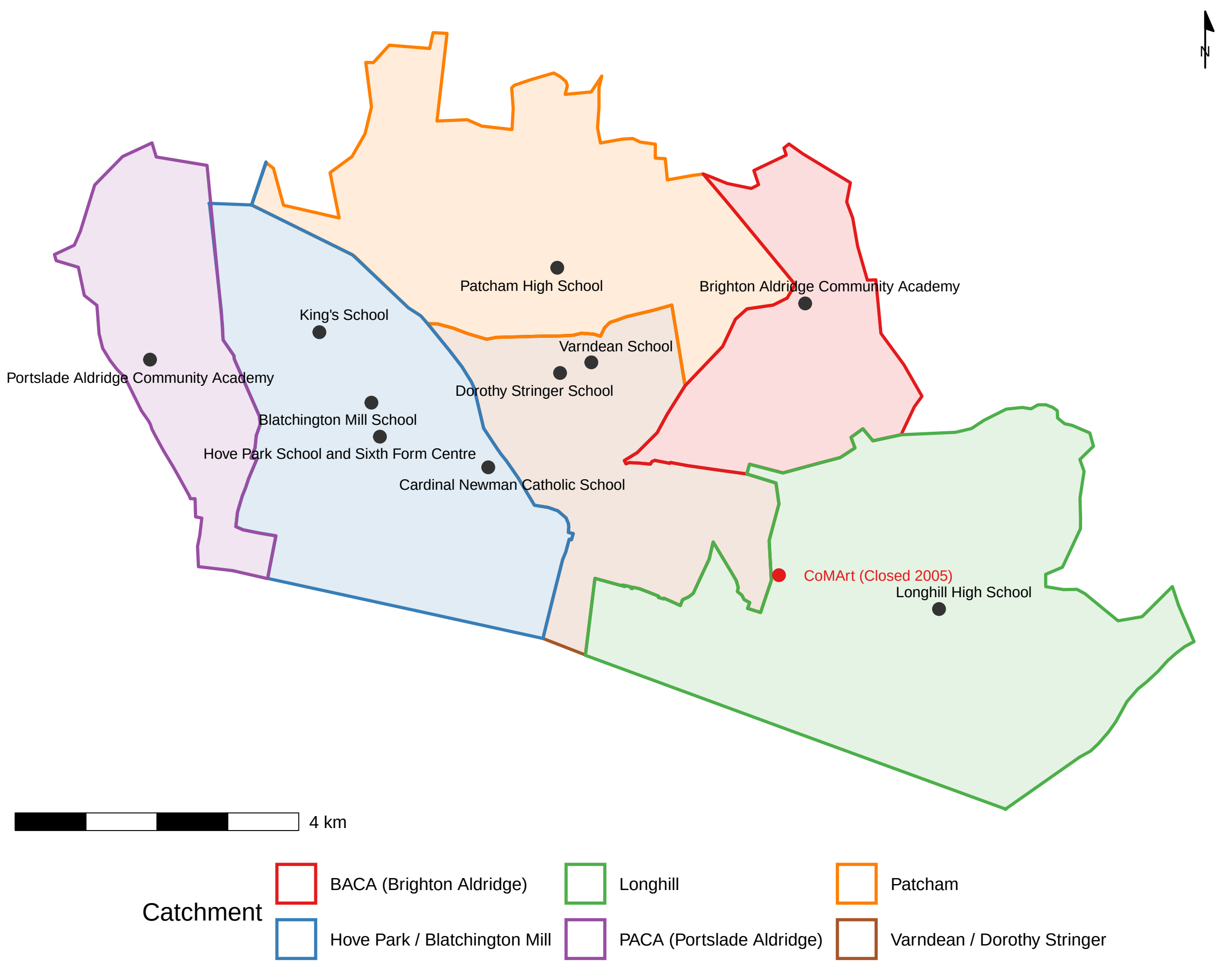

Figure 2: Brighton and Hove secondary schools with 2025/26 catchment boundaries.

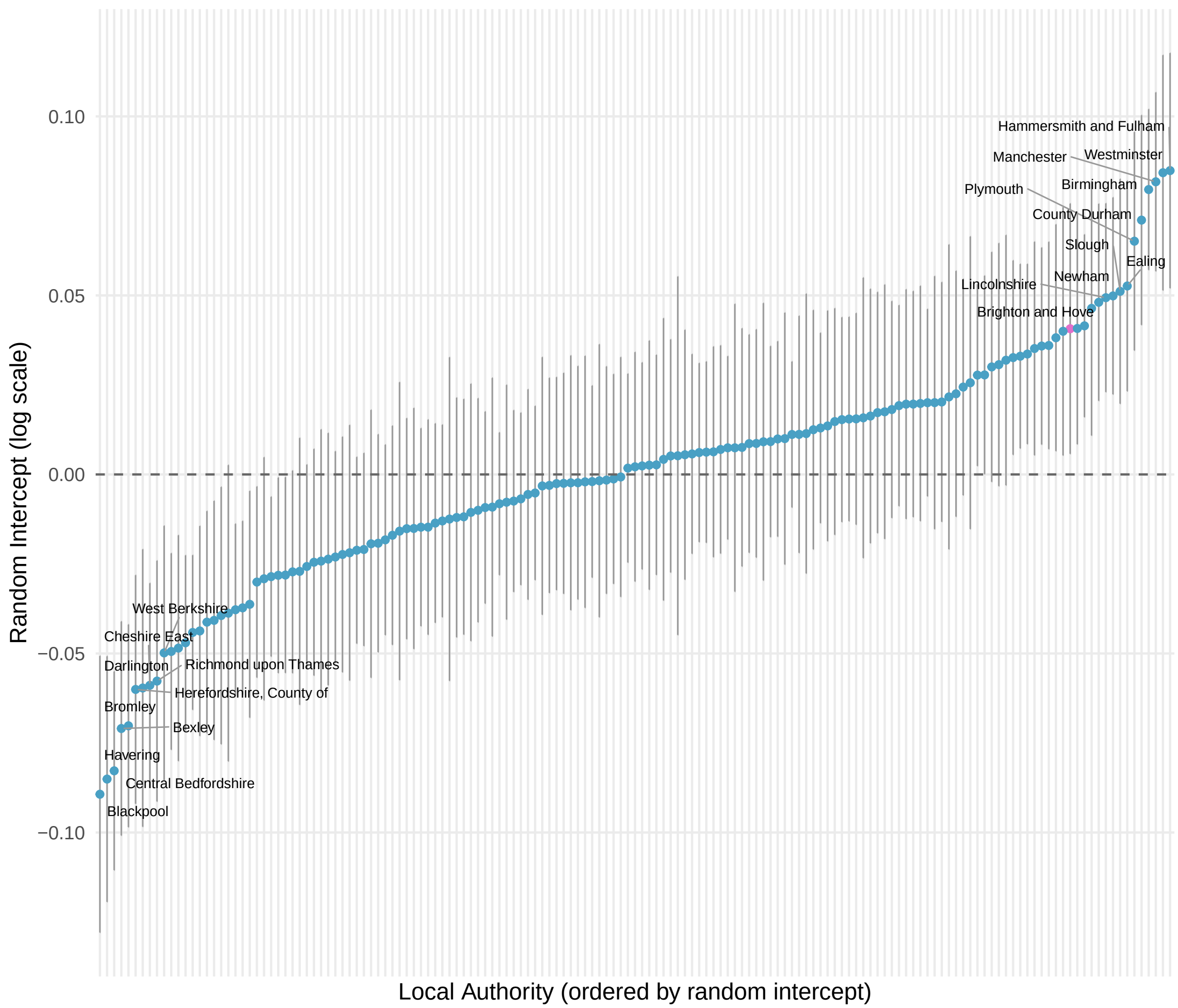

Figure 3: LA random intercepts for disadvantaged-pupil attainment (imputed full panel model). Top and bottom 10 LAs labelled; Brighton and Hove highlighted.

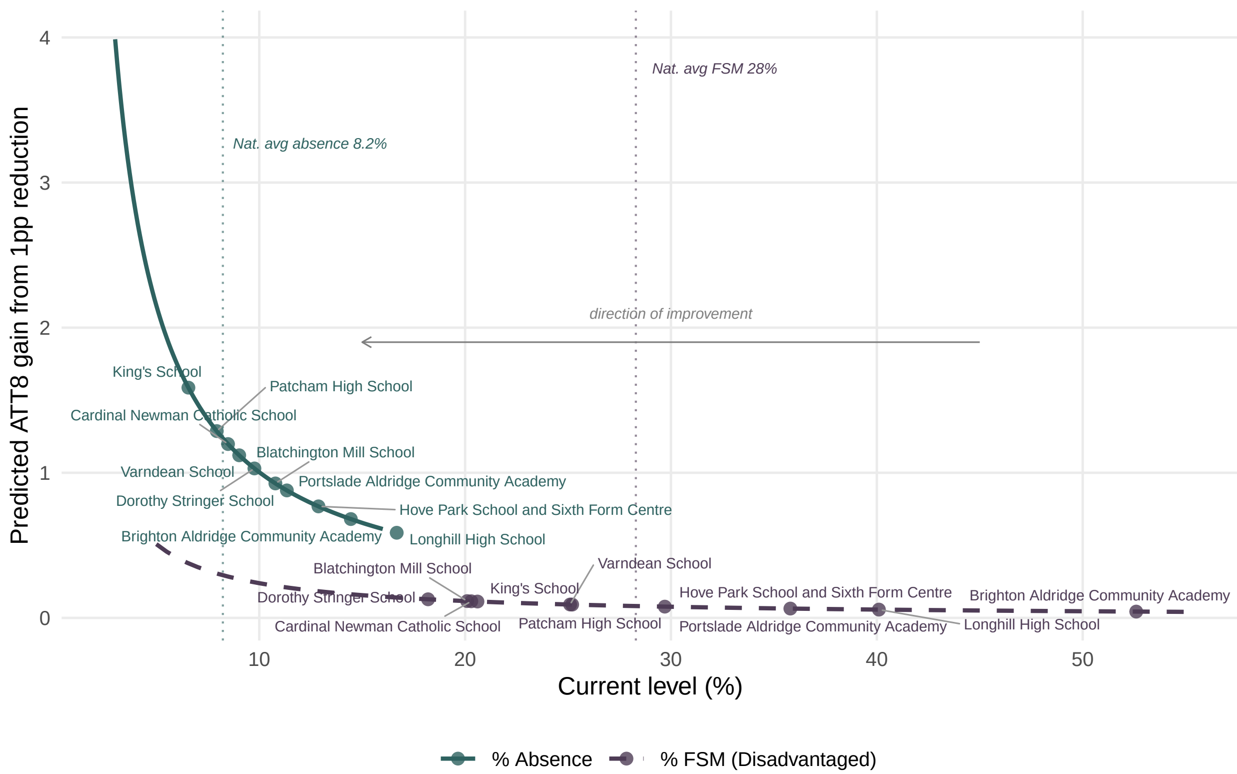

Figure 4: Accelerating returns: each percentage-point reduction buys a progressively larger ATT8 gain (reading right to left). Absence delivers substantially larger gains than FSM at every level. Brighton and Hove schools marked on each curve.

Table 3: Value-added rankings for Brighton and Hove schools (panel model residuals by year, in Attainment 8 points).

Rank	School	2021-22	2022-23	2023-24	2024-25	Mean
All Pupils						
1	Dorothy Stringer School	+2.5	+3.1	+2.2	+4.8	**+3.1**
2	Varndean School	+2.7	+2.6	+5.3	+0.5	**+2.8**
3	King's School	+3.5	+0.9	+5.0	-0.5	**+2.2**
4	Brighton Aldridge Community Academy	+2.0	+1.4	+1.2	+0.8	**+1.4**
5	Portslade Aldridge Community Academy	+3.8	+0.6	+2.3	-1.3	**+1.4**
6	Hove Park School and Sixth Form Centre	+1.5	+0.9	+1.4	+0.4	**+1.0**
7	Blatchington Mill School	-1.5	-1.0	+2.0	+2.4	**+0.5**
8	Cardinal Newman Catholic School	+1.3	-2.2	+1.3	-1.5	**-0.3**
9	Longhill High School	-0.2	-2.3	-1.1	-2.5	**-1.5**
10	Patcham High School	-2.7	-2.8	-4.4	-2.9	**-3.2**
Disadvantaged Pupils						
1	Brighton Aldridge Community Academy	+3.1	-1.3	+4.6	+5.7	**+3.1**
2	Varndean School	+2.2	+3.2	+3.9	-2.4	**+1.7**
3	Cardinal Newman Catholic School	+2.9	-5.4	+6.5	+2.3	**+1.6**
4	Dorothy Stringer School	+2.2	+2.6	-0.2	+1.4	**+1.5**
5	Blatchington Mill School	-5.0	+1.1	+4.7	+2.9	**+0.9**
6	King's School	+4.8	+1.4	+3.8	-6.2	**+0.9**
7	Hove Park School and Sixth Form Centre	+3.0	-2.5	+2.0	+1.1	**+0.9**
8	Portslade Aldridge Community Academy	+3.2	-3.4	+4.1	-1.9	**+0.5**
9	Patcham High School	+4.1	+1.5	-11.6	-3.5	**-2.4**
10	Longhill High School	-5.1	-6.4	+1.4	-0.9	**-2.7**

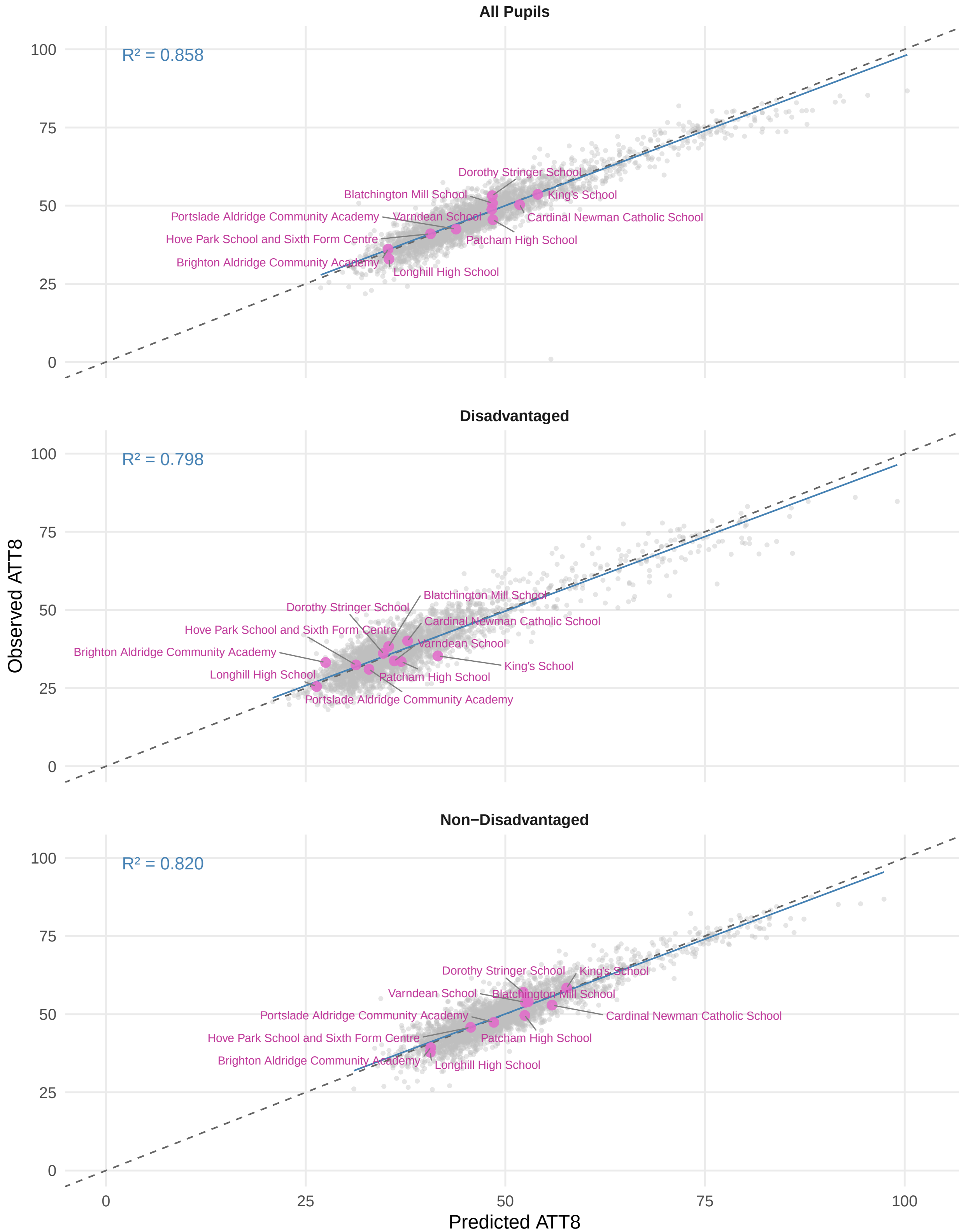

Figure 5: Observed versus predicted Attainment 8 for Brighton and Hove schools (highlighted) against all schools nationally, 2024–25.

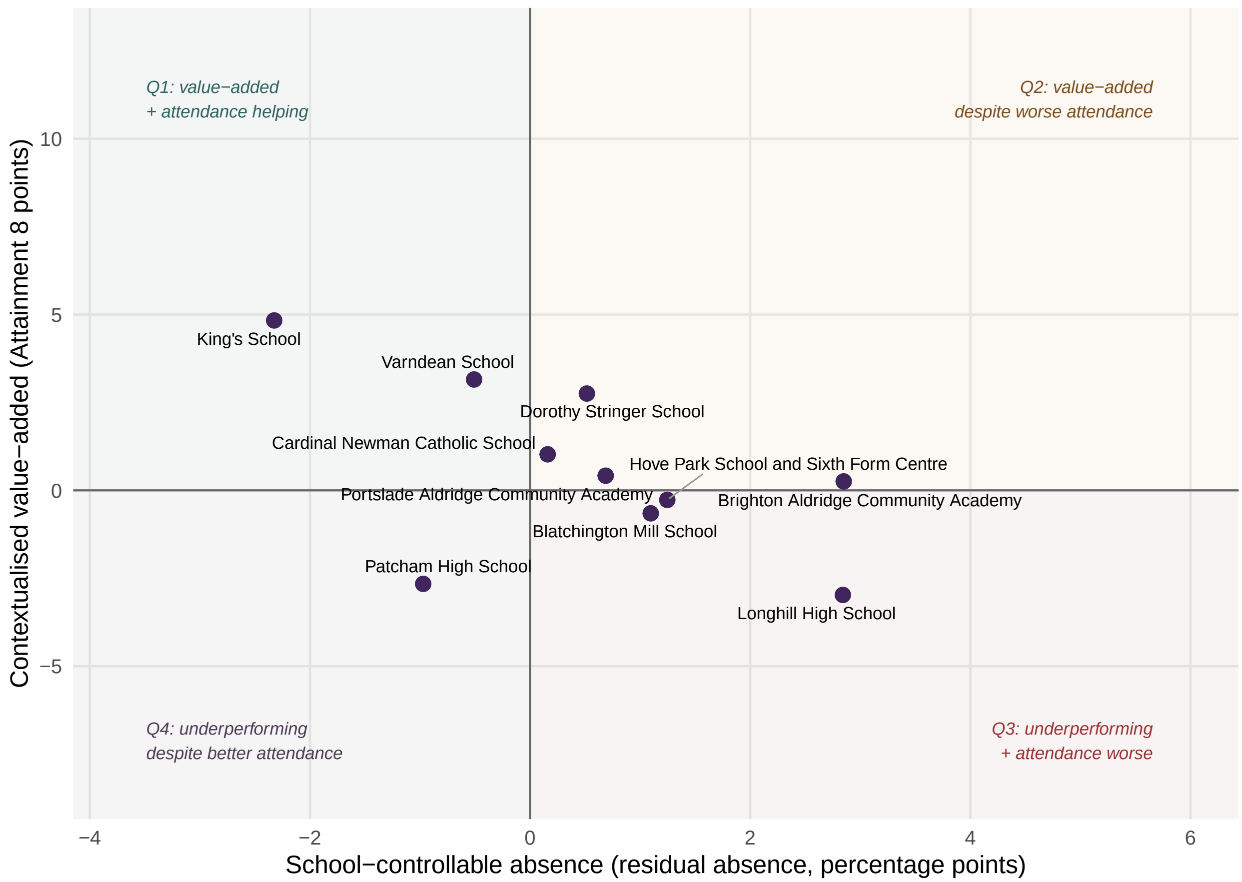

Figure 6: Brighton and Hove secondaries on the joint-signal plane: contextualised value-added (vertical, in Attainment 8 points) versus the school-controllable absence component (horizontal, in percentage points). The reference lines at zero on each axis split the plane into four narrative quadrants discussed in the text. The full national version with a local-authority filter is available interactively in the project's HTML report.